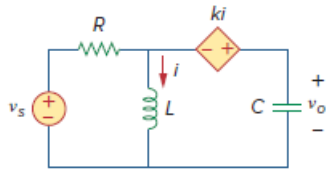


Problem

Using Fig. 16.81, design a problem to help other students better understand circuit analysis in the s -domain with circuits that have dependent sources.

Figure 16.81:



Step-by-step solution

Step 1 of 1

Using Fig. 16.53, design a problem to help other students better understand circuit analysis in the s-domain with circuits that have dependent sources.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

Problem

In the circuit of Fig. 16.53, let $i(0) = 1$ A, $v_o(0) = 2$ V, and $v_s = 4e^{-2t}u(t)$ V. Find $v_o(t)$ for $t > 0$.

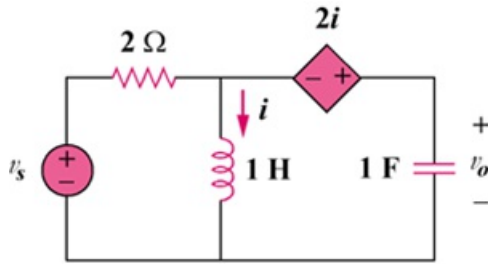
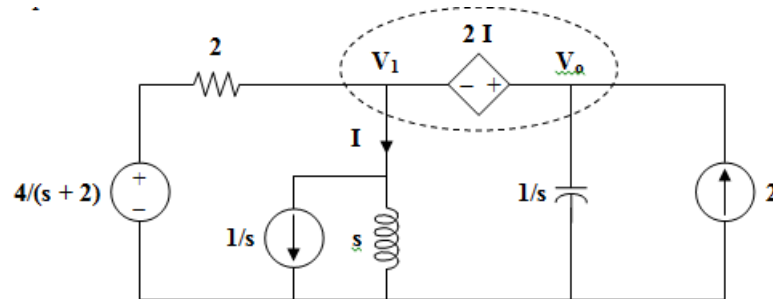


Figure 16.53

Solution

We incorporate the initial conditions in the s-domain circuit as shown below.



At the supernode,

$$\frac{(4/(s+2)) - V_1}{2} + 2 = \frac{V_1}{s} + \frac{1}{s} + sV_o$$

$$\frac{2}{s+2} + 2 = \left(\frac{1}{2} + \frac{1}{s}\right)V_1 + \frac{1}{s} + sV_o \quad (1)$$

But $V_o = V_1 + 2I$ and $I = \frac{V_1 + 1}{s}$

$$V_o = V_1 + \frac{2(V_1 + 1)}{s} \longrightarrow V_1 = \frac{V_o - 2/s}{(s+2)/s} = \frac{sV_o - 2}{s+2} \quad (2)$$

Substituting (2) into (1)

$$\frac{2}{s+2} + 2 - \frac{1}{s} = \left(\frac{s+2}{2s}\right) \left[\left(\frac{s}{s+2}\right)V_o - \frac{2}{s+2} \right] + sV_o$$

$$\frac{2}{s+2} + 2 - \frac{1}{s} + \frac{1}{s} = \left[\left(\frac{1}{2}\right) + s \right] V_o$$

$$\frac{2s+4+2}{(s+2)} = \frac{2s+6}{s+2} = (s+1/2)V_o$$

$$V_o = \frac{2s+6}{(s+2)(s+1/2)} = \frac{A}{s+1/2} + \frac{B}{s+2}$$

$$A = (-1+6)/(-0.5+2) = 3.333, \quad B = (-4+6)/(-2+1/2) = -1.3333$$

$$V_o = \frac{3.333}{s+1/2} - \frac{1.3333}{s+2}$$

Therefore,

$$v_o(t) = \underline{(3.333e^{-t/2} - 1.3333e^{-2t})u(t) \text{ V}}$$